

Project Structure :

```
login-automation-Test/  
├── DriverSetup.java  
├── LoginActions.java  
├── LoginScenarios.java  
└── TestRunner.java
```

Driver Setup Class (DriverSetup.java)

```
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.chrome.ChromeDriver;  
  
public class DriverSetup {  
  
    public static WebDriver driver;  
  
    public static void startBrowser() {  
        driver = new ChromeDriver();  
        driver.manage().window().maximize();  
        driver.get("https://example.com/login");  
    }  
  
    public static void closeBrowser() {  
        driver.quit();  
    }  
}
```

Login Actions Class (Reusable Methods - Contains only element actions – no test logic.

) (LoginActions.java)

```
import org.openqa.selenium.By;  
import org.openqa.selenium.WebDriver;  
  
public class LoginActions {  
  
    WebDriver driver;  
  
    public LoginActions(WebDriver driver) {  
        this.driver = driver;  
    }  
}
```

```
}

public void enterUsername(String username) {
    driver.findElement(By.id("username")).clear();
    driver.findElement(By.id("username")).sendKeys(username);
}

public void enterPassword(String password) {
    driver.findElement(By.id("password")).clear();
    driver.findElement(By.id("password")).sendKeys(password);
}

public void clickLogin() {
    driver.findElement(By.id("loginBtn")).click();
}

public String getErrorMessage() {
    return driver.findElement(By.id("error")).getText();
}

public boolean isDashboardDisplayed() {
    return driver.findElement(By.id("dashboard")).isDisplayed();
}

public void login(String username, String password) {
    enterUsername(username);
    enterPassword(password);
    clickLogin();
}
}
```

All Positive + Negative Scenarios Class (This is where real testing logic lives)
(LoginScenarios.java)

```
public class LoginScenarios {

    LoginActions login;

    public LoginScenarios(LoginActions login) {
        this.login = login;
    }

    // Positive Scenario
```

```
public void validLogin() {
    login.login("validUser", "validPass");

    if (login.isDashboardDisplayed()) {
        System.out.println("PASS → Valid Login");
    } else {
        System.out.println("FAIL → Valid Login");
    }
}

// Invalid username
public void invalidUsername() {
    login.login("wrongUser", "validPass");

    if (login.getErrorMessage().equals("Invalid credentials")) {
        System.out.println("PASS → Invalid Username");
    } else {
        System.out.println("FAIL → Invalid Username");
    }
}

// Invalid password
public void invalidPassword() {
    login.login("validUser", "wrongPass");

    if (login.getErrorMessage().equals("Invalid credentials")) {
        System.out.println("PASS → Invalid Password");
    } else {
        System.out.println("FAIL → Invalid Password");
    }
}

// Empty username
public void emptyUsername() {
    login.login("", "password");

    if (login.getErrorMessage().equals("Username required")) {
        System.out.println("PASS → Empty Username");
    } else {
        System.out.println("FAIL → Empty Username");
    }
}

// Empty password
public void emptyPassword() {
    login.login("username", "");
}
```

```
        if (login.getErrorMessage().equals("Password required")) {
            System.out.println("PASS → Empty Password");
        } else {
            System.out.println("FAIL → Empty Password");
        }
    }

    // Empty fields
    public void emptyFields() {
        login.login("", "");

        if (login.getErrorMessage().equals("All fields required")) {
            System.out.println("PASS → Empty Fields");
        } else {
            System.out.println("FAIL → Empty Fields");
        }
    }

    // Locked account
    public void lockedUser() {
        login.login("lockedUser", "password");

        if (login.getErrorMessage().equals("User locked")) {
            System.out.println("PASS → Locked User");
        } else {
            System.out.println("FAIL → Locked User");
        }
    }
}
```

Test Runner (Executable Program - TestRunner.java)

```
public class TestRunner {

    public static void main(String[] args) {

        DriverSetup.startBrowser();

        LoginActions actions = new LoginActions(DriverSetup.driver);
        LoginScenarios tests = new LoginScenarios(actions);

        tests.validLogin();
        tests.invalidUsername();
        tests.invalidPassword();
    }
}
```

```
tests.emptyUsername();  
tests.emptyPassword();  
tests.emptyFields();  
tests.sqlInjectionAttempt();  
tests.lockedUser();  
  
DriverSetup.closeBrowser();  
}  
}
```

